2. 1. 2020

Tutorial I

Q1. Using axioms of probability, prove
(i) $P(\bar{E}) = 1 - P(E)$ (ii) If $E \subseteq F$, $P(E) \leq P(F)$

Q2. If A and B are any events, not necessarily mutually exclusive, then
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Q3. Show that
$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(CA) + P(ABC)$$

Q4. For non-negative integers $n_1, n_2, \ldots, n_r$ summing to n,
$$\binom{n}{n_1, n_2, \ldots n_r} = \frac{n!}{n_1!\, n_2! \cdots n_r!}$$
is the number of divisions of n items into r distinct non-overlapping subgroups of sizes $n_1, n_2, \ldots, n_r$.

Using this result find the prob. that one of the players receives all the 13 spades when the entire deck of 52 cards is dealt out to 4 players.

Q5. A total of 36 members of a club play tennis (T), 28 play squash (S) and 18 play badminton (B). Furthermore, 22 of the members play both T and S, 12 play both T and B, 9 play both S and B, and 4 play all three sports.

How many members of this club play atleast one of these sports? Let N denote the total number of members of the club.

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TUTORIAL II

Q1. For events $A_1, A_2, \ldots, A_n$, show that

$$P[A_1 \cap A_2 \cap \cdots \cap A_n] = P(A_1)\, P(A_2|A_1) \cdot P[A_3|A_1 \cap A_2) \cdots$$
$$\cdots P(A_n | A_1 \cap A_2 \cdots \cap A_{n-1})$$

provided all the probabilities on right are defined.

Q2. Suppose that a survey of 100 computer installations in a certain city shows that 75 of them have atleast one brand X computer. If three of these installations are chosen at random without replacement, what is the probability that each of them has atleast one brand X machine?

Q3. Suppose independent trials of a random experiment are performed repeatedly. Each trial results in one and only one of three outcomes 1, 2, 3. Assume that the probability of getting outcome i on any given trial is p_i, such that $p_1 + p_2 + p_3 = 1$. What is the probability that outcome 1 is observed before outcome 2?

Q4. Let E_i $1 \le i \le n$ be n independent events. Show that

$$P\left(\bigcup_{i=1}^{n} E_i\right) = 1 - \prod_{i=1}^{n} (1 - P(E_i)).$$

Q5. Find the probabilities of randomly drawing two aces in succession from an ordinary deck of 52 playing cards, if we sample

(a) without replacement (b) with replacement.

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Tutorial III

Q1. If n people are present in a room, what is the probability that no two of the n people have the same birthday. (Ignore the possibility of someone having been born on Feb 29).

Q2 Four coins are flipped. If the outcomes are assumed independent, what is the probability that two heads and two tails are obtained?

Q3 Consider two urns. The first contains 2 white and 7 black balls, and the second contains 5 white and 6 black balls. We flip a fair coin and then draw a ball from the first urn or the second urn depending on whether the outcome was heads or tails. What is the conditional probability that the outcome of the toss was heads given that a white ball was selected?

Q4 (Random Coin Tossing)

An urn contains n coins. When the i th coin is tossed, it turns up heads with probability p_i, $1 \le i \le n$. (i) Suppose a coin is picked at random from the urn and tossed. What is the probability of getting a head?

(ii) Suppose a coin is picked randomly from the urn and we do not know which it is. What is the probability that it is the ith coin given that it landed heads?

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TUTORIAL IV

Q1. It is known that any item produced by a certain machine will be defective with probability 0.1, independently of any other item. In a sample of three items, what is the probability that at most one will be defective?

Q2. Suppose that an airplane engine will fail, when in flight, with probability $1-p$ independently from engine to engine; suppose that airplane will make successful flight if atleast 50 percent of engines remain operative. For what values of p is a four engine plane preferable to a two engine plane.

Q3. Suppose that independent trials, each of which results in any of m possible outcomes with respective probabilities $p_1, p_2, \ldots, p_m$, $\sum_{i=1} p_i = 1$ are continually performed. Let X denote the number of trials needed until each outcome has occurred atleast once. Find (i) $P(X=n)$. (ii) $P(X>n)$

Q4 The cdf of a discrete r.v X is given by

$$F(a) = \begin{cases} 0 & a<1 \\ \frac{1}{2} & 1 \le a < 2 \\ \frac{5}{6} & 2 \le a < 3 \\ 1 & 3 \le a \end{cases}$$

Draw graph of $F(x)$ and find pmf of X

Q5 X is a continuous r.v. with pdf

$$f(x) = \begin{cases} c\, e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find c (ii) Sketch cdf.

(iii) $P(X \le \frac{1}{2})$

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30.1.2020
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Tutorial V

Q1. Suppose that a r.v. X has a discrete distribution with the following pmf:

$$f(x) = \begin{cases} \dfrac{c}{2^x}, & x = 0, 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

Find the value of c.

Q2. Let X be a discrete r.v with pmf

$$p(x) = c(x+1)(8-x), \quad x = 0, 1, 2, \ldots 7.$$

(i) Find c (ii) $P(X \geq 5)$ (iii) $E[X]$.

Q3. Show that there does not exist any number c such that the following would be a pmf,

$$f(x) = \begin{cases} c/x & \text{for } x = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

Q.4. If X is uniformly distributed over $(0, 10)$, calculate the probability that (i) $X < 3$ (ii) $X > 7$ (iii) $1 < X < 6$

Q5. Calculate the expectation of a geometric r.v with pmf

$$p(n) = p(1-p)^{n-1}, \quad n = 1, 2, \ldots$$

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TUTORIAL VI

4. 2. 2020
(Tuesday)

Q1. Suppose that the cdf of the rv X is given by

$$F_X(x) = 1 - e^{-x^2}, \quad x > 0$$

Evaluate (i) $P(X > 1.2)$ (ii) $P(1 < X < 3)$
(iii) $E(X)$ (iv) $V(X)$

Q2. The r.v X has pdf

$$f(x) = \begin{cases} ax + bx^2 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If $E(X) = 0.6$, find (a) $P(X < \frac{1}{2})$
(b) Variance X.

Q3. If X is a binomial r.v. such that
$E(X) = 6$ and Var $X = 2.4$,
find $P(X = 5)$

Q4. Let X be a random variable having expected value μ and variance σ^2. Find
(i) $E\left[\frac{X-\mu}{\sigma}\right]$ (ii) $E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] - \left[E\left(\frac{X-\mu}{\sigma}\right)\right]^2$

Q5. Plot pdfs of the Gaussian random variables:
$X_1 \sim N(\mu, \sigma^2)$ and $X_2 \sim N(\mu, 4\sigma^2)$

$$f_1(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-(x-\mu)^2/2\sigma^2\right)$$

$$f_2(x) = \frac{1}{\sqrt{2\pi}\,(2\sigma)} \exp\left(-(x-\mu)^2/2\cdot(2\sigma)^2\right),$$

$$-\infty < x < \infty.$$

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Tutorial VII

Q1. Suppose that X has a continuous distribution for which the pdf is as follows:

$$f(x) = \begin{cases} 4x^3 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the mean and median.

Q2. Suppose that X has a continuous distribution for which the pdf is as follows:

$$f(x) = \begin{cases} 1/2 & \text{for } 0 \le x \le 1 \\ 1 & \text{for } 2.5 \le x \le 3 \\ 0 & \text{otherwise} \end{cases}$$

Determine the median and comment on the result.

Q3. Let X be a rv with finite variance σ^2 and let $\mu = E(X)$. Show that for any number d,

$$E[(X-\mu)^2] \le E[(X-d)^2].$$

There will be equality iff $d = \mu$.

<u>Note</u>: The number $E[(X-d)^2]$ is called the mean square error (MSE) of the prediction d.

Q4 An analog signal X at a detector may be modeled as a Gaussian rv $N(200, 256)$ at a fixed point in time. Find $P(X > 240)$. Also determine $P(X > 240 \mid X > 210)$.

Q5. The lifetime of a component follows Weibull pdf. with shape parameter $\alpha = 2$. The cdf is

$$F_X(x) = 1 - e^{-\lambda x^2}$$

We are given $P(X < 100 \mid X > 90) = 0.15$.

How will you determine λ?

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